

NOVA Voice Assistant – QA Setup Guide

Before running the project, please complete the following setup steps.

1. Install Python

- Install **Python 3.12.x**.
- Ensure Python is added to the system **PATH** during installation.

Note: The project is developed using Python 3.12. Other versions may cause compatibility issues with the voice libraries.

2. Install Required Python Packages

Open the project directory in Command Prompt and install all the required Python packages.

- asyncio
 - screen_brightness_control
 - SpeechRecognition
 - playsound
 - edge_tts
 - mysql_connector_python
 - PyAudio
 - Playsound
 - pyttsx3
 - Plyer
 - requests
-

3. Configure MySQL Database

- Install and start **MySQL Server**.
- Create a MySQL database named:

```
CREATE DATABASE assistant;
```

```
USE assistant;
```

Table 1: `schedule_task`

```
CREATE TABLE schedule_task (
```

```
id INT AUTO_INCREMENT PRIMARY KEY,  
  
task_name VARCHAR(255) NOT NULL,  
  
task_date DATE NOT NULL,  
  
task_time TIME NOT NULL,  
  
status VARCHAR(20) DEFAULT 'Pending'  
  
);
```

This table is used by the **Reminder Module** to store scheduled reminders.

Table 2: **logs**

```
CREATE TABLE logs (  
  
id INT AUTO_INCREMENT PRIMARY KEY,  
  
module VARCHAR(100),  
  
user_command TEXT,  
  
status VARCHAR(50),  
  
log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

This table stores the execution history of user commands for the **Logs Module**.

Update Database Credentials

Before running the project, update the following details in the [config.py](#) file:

- Host
 - Username
 - Password
 - Database Name (**assistant**)
-

4. Check Configuration File

Before running the project, review the **config.py** file.

- Verify the project configuration.
 - Refer to the list of supported voice commands available in the file.
 - The QA team should use these commands while testing the application.
-

5. Run the Project

Navigate to the project **SRC** folder and execute:

python [main.py](#)

6. Common Issues

Microphone not detected

- Check Windows microphone permissions.
- Ensure the correct microphone is selected.

MySQL Connection Error

- Verify MySQL Server is running.
- Check username and password.
- Ensure the **assistant** database exists.

ModuleNotFoundError

Install the missing package.

Example:

```
pip install package_name
```

Voice Not Speaking

- Check Internet connection.
- Verify the selected voice in the settings.
- Ensure the Edge-TTS package is installed.

Command Not Recognized

- Speak clearly.

- Reduce background noise.
 - Ensure the microphone is working correctly.
-

7.Common Voice Commands

NOVA supports simple natural language voice commands for performing different tasks. Some commonly used commands are:

- **Open Application:** "Open Chrome"
- **Open Website:** "Open YouTube"
- **Web Search:** "Search for Python programming"
- **Open File:** "Open file report.pdf"
- **Open Folder:** "Open folder Documents"
- **Create Folder:** "Create folder NOVA"
- **Delete File:** "Delete file notes.txt"
- **Rename File:** "Rename file report.pdf to final_report.pdf"
- **Set Reminder:** "Remind me to submit the project tomorrow at 10 AM"
- **Open Settings:** "Open settings"
- **Change Language and Voice:** "Change language and voice"
- **View Logs:** "Show logs"
- **Exit/Go Back:** "Go back"